

YOUTUBE TRENDING DATA ANALYSIS & DASHBOARD

Project Report | Python, Pandas, SQLite and Tableau

1. Project Overview

This project analyzes a YouTube trending-video dataset to identify patterns in video popularity, engagement, title sentiment, publishing country, content category, and trending duration. Python and Pandas were used for data preparation and exploratory analysis, SQLite for structured storage and querying, and Tableau for interactive visualization and dashboard development.

2. Dataset Description

The dataset contains video ID, trending date, title, channel title, category ID, publish date, time frame, published day of week, publishing country, tags, views, likes, dislikes, comment count, and video-status indicators. Additional fields were created for title sentiment score and sentiment class.

Trending period: November 2017 to June 2018. **Publishing period:** July 2006 to June 2018.

3. Technologies Used

Technology	Purpose
Python	Data cleaning, transformation and exploratory analysis
Pandas	DataFrame manipulation, grouping and statistics
SQLite	Structured storage and SQL-based analysis
VADER Sentiment Analysis	Classification of titles as Positive, Neutral or Negative
Tableau	Interactive charts, filters and dashboard

4. Data Analysis Performed

- Compared views, likes and comments across categories and countries.
- Calculated the number of days each video remained in the trending dataset.
- Performed title sentiment analysis using sentiment scores and three sentiment classes.
- Compared sentiment distribution across countries and categories.
- Identified highly trending videos and examined their performance metrics.

5. Key Findings

Measure	Observed result
Title sentiment	Neutral 63.24%; Positive 20.06%; Negative 16.70%
Average views by sentiment	Negative ~3.59M; Positive ~2.45M; Neutral ~2.10M
Average views by country	GB ~5.91M; US ~2.36M; Canada ~1.15M; France ~0.42M
Average trending duration	GB ~10.95 days; US ~7.91; Canada ~1.53; France ~1.38

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6. Tableau Visualization & Dashboard

The Tableau dashboard combines category, country, sentiment and trending-duration analysis. Interactive filters allow the viewer to explore results by publishing country, category and title sentiment.

Main Visualizations

- Videos by Category — compares the number of unique videos represented in each category.
- Average Views by Country — compares average views across publishing countries.
- Videos by Publishing Country — shows video volume by country.
- Title Sentiment Distribution — displays neutral, positive and negative title proportions.
- Average Views by Title Sentiment — compares performance across sentiment groups.
- Average Trending Duration by Country — compares trending duration across countries.

7. Analytical Insights

Neutral titles form the largest share of the analyzed dataset. In the calculated results, negative-sentiment titles had the highest average views, likes and comments, followed by positive and neutral titles. This is an association in the dataset and should not be interpreted as proof that title sentiment causes higher views.

Country-level analysis shows differences in both content volume and average performance. France and Canada account for larger numbers of videos in the analyzed country grouping, while GB and the US show higher average views. Trending-duration analysis also shows substantial differences between countries.

8. Project Outcome

The project demonstrates an end-to-end analytics workflow: understanding and preparing a dataset, creating analytical variables, performing statistical and sentiment analysis, storing data in SQLite, and communicating results through Tableau.

9. Conclusion

The YouTube Trending Analysis project provides a practical example of combining Python-based analysis with business-intelligence visualization. The final dashboard makes patterns in popularity, engagement, sentiment and trending behavior easier to explore and communicate.

10. Future Scope

- Add time-series analysis of trending behavior.
- Use category names instead of category IDs for readability.
- Develop predictive models for views or trending duration.
- Add channel-level and engagement-rate analysis.
- Expand dashboard drill-downs and interactive tooltips.

Project Repository: [youtube-trending-analysis-dashboard](#)